

7.7 – Roots and Factors Practice

Algebra 2

Name: _____

Period: _____

Section A: Roots $\leftrightarrow$ Factors

Given the roots, write the polynomial in factored form.

1. Roots: $x = 2, x = -3, x = 5$ $f(x) = \underline{\hspace{2cm}}$	2. Roots: $x = 0, x = 4, x = -1$ $f(x) = \underline{\hspace{2cm}}$
3. Roots: $x = 1, x = 1, x = -4$ $f(x) = \underline{\hspace{2cm}}$	4. Roots: $x = -2, x = 3, x = 6$ $f(x) = \underline{\hspace{2cm}}$

Given the factored form, list all the roots.

5. $f(x) = (x - 4)(x + 2)(x - 1)$ Roots: $x = \underline{\hspace{2cm}}$	6. $g(x) = x(x + 5)(x - 3)$ Roots: $x = \underline{\hspace{2cm}}$
7. $h(x) = (x + 1)(x + 1)(x - 2)$ Roots: $x = \underline{\hspace{2cm}}$	8. $p(x) = 2(x - 3)(x + 4)(x - 1)$ Roots: $x = \underline{\hspace{2cm}}$

Section B: Verify Roots with Synthetic Division

Use synthetic division to determine if the given value is a root. (Remainder = 0 means YES!)

9. Is $x = 2$ a root of $x^3 - 7x + 6$? $\begin{array}{r rrrr} +2 & 1 & 0 & -7 & 6 \\ \hline \end{array}$ Remainder: _____ Root? YES / NO	10. Is $x = 3$ a root of $x^3 - 7x + 6$? $\begin{array}{r rrrr} +3 & 1 & 0 & -7 & 6 \\ \hline \end{array}$ Remainder: _____ Root? YES / NO
11. Is $x = 4$ a root of $x^3 - 3x^2 - 10x + 24$? $\begin{array}{r rrrr} +4 & 1 & -3 & -10 & 24 \\ \hline \end{array}$ Remainder: _____ Root? YES / NO	12. Is $x = -2$ a root of $x^3 + 3x^2 - x - 3$? $\begin{array}{r rrrr} -2 & 1 & 3 & -1 & -3 \\ \hline \end{array}$ Remainder: _____ Root? YES / NO

Section C: Factor Completely

Use the given root to factor completely. Show synthetic division, then factor the quadratic.

13. $f(x) = x^3 - 2x^2 - 5x + 6$ (Given: $x = 1$ is a root)

Quotient: _____ Factor the quotient: (____)(____)

Completely factored: $f(x) = \underline{\hspace{2cm}}$

14. $g(x) = x^3 + 4x^2 - 7x - 10$ (Given: $x = 2$ is a root)

Quotient: _____ Factor the quotient: (____)(____)

Completely factored: $g(x) =$ _____

15. $h(x) = x^3 + 6x^2 + 11x + 6$ (Given: $x = -1$ is a root)

Quotient: _____ Factor the quotient: (____)(____)

Completely factored: $h(x) =$ _____

16. $p(x) = x^3 + 2x^2 - 9x - 18$ (Given: $x = 3$ is a root)

Quotient: _____ Factor the quotient: (____)(____)

Completely factored: $p(x) =$ _____

17. $q(x) = 2x^3 - x^2 - 13x - 6$ (Given: $x = 3$ is a root)

Quotient: _____ Factor the quotient: (____)(____)

Completely factored: $q(x) =$ _____

Section D: Identify X-Intercepts from a Table

Use the table to identify the x -intercepts, roots, and linear factors of the polynomial.

18. The table shows values for $f(x) = x^3 - 4x^2 - 11x + 30$.

x	-4	-3	-2	-1	0	1	2	3	4	5	6
$f(x)$	-54	0	28	36	30	16	0	-12	-14	0	36

X-intercepts: _____

Roots: _____

Linear Factors: _____

Section D: Factor and Find All Roots

Factor completely using grouping or special patterns. List all roots, then plot them on the x -axis of the grid.

19. $f(x) = x^3 + 3x^2 - 25x - 75$

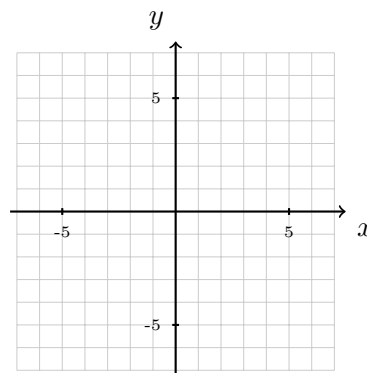
Factor by Grouping

Show your work:

Plot the roots on the graph. Use a calculator to complete the graph:

Factored form: $f(x) =$

Roots: $x =$ _____



20. $g(x) = x^3 - 27$

Difference of Cubes

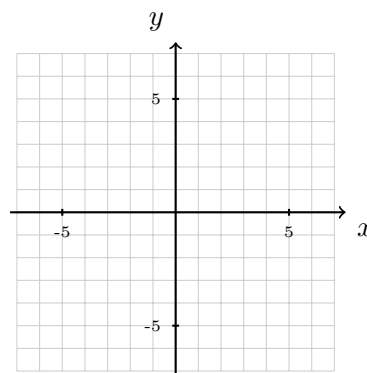
(Hint: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$)

Show your work:

Plot the roots on the graph. Use a calculator to complete the graph:

Factored form: $g(x) =$

Roots: $x =$ _____



21. $h(x) = x^4 - 10x^2 + 9$

Quadratic in Form (Degree 4)

(Hint: Let $u = x^2$. Rewrite as $u^2 - 10u + 9$, then factor and substitute back.)

Show your work:

Plot the roots on the graph:

Factored form: $h(x) =$

Roots: $x =$ _____

